

# Business Problem

## Import Libraries

```
import numpy as np import matplotlib.pyplot as plt import pandas as pd
```

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Load Dataset

**path = r"**<https://drive.google.com/uc>

```
In [2]: df = pd.read_csv(r"D:\Classwork\student_info.csv")
```

```
In [3]: df
```

```
Out[3]:
```

|     | study_hours | student_marks |
|-----|-------------|---------------|
| 0   | 6.83        | 78.50         |
| 1   | 6.56        | 76.74         |
| 2   | NaN         | 78.68         |
| 3   | 5.67        | 71.82         |
| 4   | 8.67        | 84.19         |
| ... | ...         | ...           |
| 195 | 7.53        | 81.67         |
| 196 | 8.56        | 84.68         |
| 197 | 8.94        | 86.75         |
| 198 | 6.60        | 78.05         |
| 199 | 8.35        | 83.50         |

200 rows × 2 columns

```
In [4]: df.shape
```

```
Out[4]: (200, 2)
```

Discover and Visualize the data to gain Insights

```
In [5]: df.info()
```

```

<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  -
0    study_hours     195 non-null    float64
1    student_marks   200 non-null    float64
dtypes: float64(2)
memory usage: 3.3 KB

```

In [ ]:

Descriptive Statistics

In [6]: `df.describe()`

Out[6]:

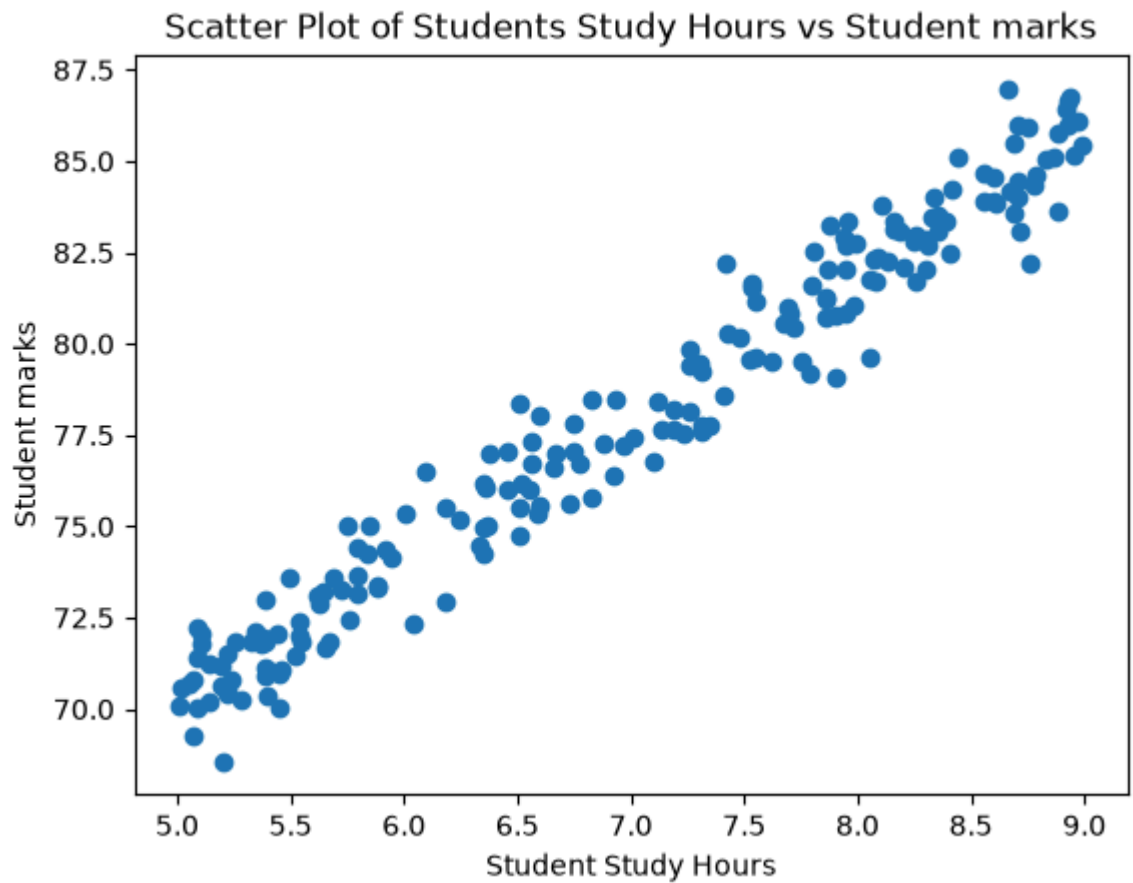
|              | study_hours | student_marks |
|--------------|-------------|---------------|
| <b>count</b> | 195.000000  | 200.000000    |
| <b>mean</b>  | 6.995949    | 77.93375      |
| <b>std</b>   | 1.253060    | 4.92570       |
| <b>min</b>   | 5.010000    | 68.57000      |
| <b>25%</b>   | 5.775000    | 73.38500      |
| <b>50%</b>   | 7.120000    | 77.71000      |
| <b>75%</b>   | 8.085000    | 82.32000      |
| <b>max</b>   | 8.990000    | 86.99000      |

In [7]:

```

plt.scatter(x= df.study_hours, y = df.student_marks)
plt.title('Scatter Plot of Students Study Hours vs Student marks')
plt.xlabel('Student Study Hours')
plt.ylabel('Student marks')
plt.show()

```



Prepare the data for Machine Learning algorithms

In [ ]: `#data cleaning`

In [9]: `df`

Out[9]:

|     | study_hours | student_marks |
|-----|-------------|---------------|
| 0   | 6.83        | 78.50         |
| 1   | 6.56        | 76.74         |
| 2   | NaN         | 78.68         |
| 3   | 5.67        | 71.82         |
| 4   | 8.67        | 84.19         |
| ... | ...         | ...           |
| 195 | 7.53        | 81.67         |
| 196 | 8.56        | 84.68         |
| 197 | 8.94        | 86.75         |
| 198 | 6.60        | 78.05         |
| 199 | 8.35        | 83.50         |

200 rows × 2 columns

```
In [10]: df.isnull().sum()
```

```
Out[10]: study_hours      5
student_marks    0
dtype: int64
```

```
In [11]: df.mean()
```

```
Out[11]: study_hours      6.995949
student_marks    77.933750
dtype: float64
```

```
In [12]: df2 = df.fillna(df.mean())
```

```
In [13]: df2.isnull().sum()
```

```
Out[13]: study_hours      0
student_marks    0
dtype: int64
```

```
In [14]: df2.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   study_hours     200 non-null   float64
1   student_marks   200 non-null   float64
dtypes: float64(2)
memory usage: 3.3 KB
```

```
In [15]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   study_hours     195 non-null   float64
1   student_marks   200 non-null   float64
dtypes: float64(2)
memory usage: 3.3 KB
```

```
In [16]: # Split Dataset
```

```
In [17]: X = df2.drop("student_marks", axis = "columns")
y = df2.drop("study_hours", axis = "columns")
print("shape of X = ", X.shape)
print("shape of y = ", y.shape)
```

```
shape of X = (200, 1)
shape of y = (200, 1)
```

```
In [18]: X
```

Out[18]:

| study_hours |          |
|-------------|----------|
| 0           | 6.830000 |
| 1           | 6.560000 |
| 2           | 6.995949 |
| 3           | 5.670000 |
| 4           | 8.670000 |
| ...         | ...      |
| 195         | 7.530000 |
| 196         | 8.560000 |
| 197         | 8.940000 |
| 198         | 6.600000 |
| 199         | 8.350000 |

200 rows × 1 columns

In [19]:

y

Out[19]:

| student_marks |       |
|---------------|-------|
| 0             | 78.50 |
| 1             | 76.74 |
| 2             | 78.68 |
| 3             | 71.82 |
| 4             | 84.19 |
| ...           | ...   |
| 195           | 81.67 |
| 196           | 84.68 |
| 197           | 86.75 |
| 198           | 78.05 |
| 199           | 83.50 |

200 rows × 1 columns

In [21]:

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X,y, test_size=0.2, random_s

print("shape of X_train =", X_train.shape)
print("shape of y_train =", y_train.shape)
print("shape of X_test =", X_test.shape)
print("shape of y_test =", y_test.shape)
```

```
shape of X_train = (160, 1)
shape of y_train = (160, 1)
shape of X_test = (40, 1)
shape of y_test = (40, 1)
```

```
In [22]: X_train
```

```
Out[22]:
```

|  | study_hours |
|--|-------------|
|--|-------------|

|     |      |
|-----|------|
| 134 | 6.51 |
|-----|------|

|    |      |
|----|------|
| 66 | 7.86 |
|----|------|

|    |      |
|----|------|
| 26 | 6.51 |
|----|------|

|     |      |
|-----|------|
| 113 | 7.95 |
|-----|------|

|     |      |
|-----|------|
| 168 | 7.95 |
|-----|------|

|     |     |
|-----|-----|
| ... | ... |
|-----|-----|

|    |      |
|----|------|
| 67 | 8.26 |
|----|------|

|     |      |
|-----|------|
| 192 | 8.71 |
|-----|------|

|     |      |
|-----|------|
| 117 | 8.83 |
|-----|------|

|    |      |
|----|------|
| 47 | 5.01 |
|----|------|

|     |      |
|-----|------|
| 172 | 7.35 |
|-----|------|

160 rows × 1 columns

```
In [23]: X_test.shape
```

```
Out[23]: (40, 1)
```

```
In [24]: y_train
```

Out[24]: **student\_marks**

|     |       |
|-----|-------|
| 134 | 78.39 |
| 66  | 81.25 |
| 26  | 74.75 |
| 113 | 80.86 |
| 168 | 82.68 |
| ... | ...   |
| 67  | 81.70 |
| 192 | 84.03 |
| 117 | 85.04 |
| 47  | 70.11 |
| 172 | 77.78 |

160 rows × 1 columns

In [25]: `y_test.shape`

Out[25]: (40, 1)

In [26]: `y_test`

Out[26]:

| student_marks |       |
|---------------|-------|
| 18            | 82.50 |
| 170           | 71.18 |
| 107           | 73.25 |
| 98            | 83.64 |
| 177           | 73.64 |
| 182           | 86.99 |
| 5             | 81.18 |
| 146           | 82.75 |
| 12            | 79.50 |
| 152           | 81.70 |
| 61            | 79.41 |
| 125           | 85.95 |
| 180           | 77.19 |
| 154           | 78.45 |
| 80            | 84.00 |
| 7             | 85.46 |
| 33            | 84.35 |
| 130           | 73.19 |
| 37            | 78.21 |
| 74            | 77.59 |
| 183           | 83.87 |
| 145           | 85.15 |
| 45            | 72.96 |
| 159           | 80.72 |
| 60            | 73.61 |
| 123           | 79.53 |
| 179           | 78.17 |
| 185           | 79.63 |
| 122           | 76.83 |
| 44            | 82.38 |
| 16            | 76.04 |
| 55            | 85.48 |
| 150           | 71.87 |



| student_marks |       |
|---------------|-------|
| 111           | 75.04 |
| 22            | 70.67 |
| 189           | 79.87 |
| 129           | 74.49 |
| 4             | 84.19 |
| 83            | 75.36 |
| 106           | 72.10 |

In [ ]:

Select a model and train it

In [27]:

```
#y = m*x + C
```

In [28]:

```
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
```

In [29]:

```
lr
```

Out[29]:

▼ LinearRegression ⓘ ⓘ  
► Parameters

In [30]:

```
lr.fit(X_train, y_train)
```

Out[30]:

▼ LinearRegression ⓘ ⓘ  
► Parameters  
► Fitted attributes

In [31]:

```
lr.get_params()
```

Out[31]:

```
{'copy_X': True,
 'fit_intercept': True,
 'n_jobs': None,
 'positive': False,
 'tol': 1e-06}
```

In [33]:

```
lr.coef_
```

Out[33]:

```
array([[3.93037294]])
```

In [34]:

```
lr.intercept_
```

Out[34]: array([50.45063632])

```
In [36]: m = 3.93
c = 50.44
y = m*10 + c
y
```

Out[36]: 89.74000000000001

```
In [37]: m = 3.93
c = 50.44
y = m*11 + c
y
```

Out[37]: 93.67

```
In [38]: m = 3.93
c = 50.44
y = m*9 + c
y
```

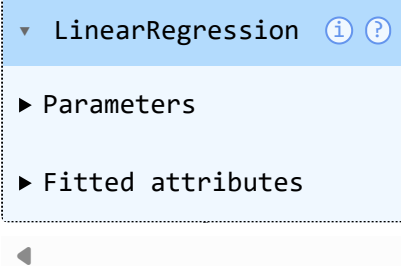
Out[38]: 85.81

```
In [39]: lr.predict([[11]]).round(2)
```

C:\Users\Amit Kumar Barnwal\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names  
warnings.warn(

Out[39]: array([[93.68]])

```
In [40]: lr
```

Out[40]: A summary of the LinearRegression object. It shows the class name 'LinearRegression' with information and help icons. Below it are two expandable sections: 'Parameters' and 'Fitted attributes', each with a right-pointing triangle icon. At the bottom of the summary box are left and right arrow icons for navigation.

```
In [41]: y_pred = lr.predict(X_test)
y_pred
```

```
Out[41]: array([[83.50507271],
               [70.84927186],
               [72.93236952],
               [85.35234799],
               [73.20749562],
               [84.48766595],
               [80.12495199],
               [81.85431608],
               [80.91102657],
               [82.20804964],
               [78.98514384],
               [84.84139951],
               [77.84533568],
               [77.68812077],
               [83.22994661],
               [85.78468901],
               [84.9593107 ],
               [72.61793968],
               [78.71001773],
               [79.18166248],
               [84.2911473 ],
               [85.6274741 ],
               [74.74034107],
               [81.3433676 ],
               [72.02838374],
               [80.40007809],
               [78.98514384],
               [82.09013845],
               [77.94732382],
               [82.24735337],
               [75.44780819],
               [84.60557713],
               [71.63534645],
               [75.48711192],
               [70.29901965],
               [78.98514384],
               [75.32989701],
               [84.52696967],
               [74.07217767],
               [71.4388278 ]])
```

```
In [46]: pd.DataFrame(np.c_[X_test, y_test, y_pred], columns = ["study_hours", "student_m
```

Out[46]:

|           | <b>study_hours</b> | <b>student_marks_original</b> | <b>student_marks_predicted</b> |
|-----------|--------------------|-------------------------------|--------------------------------|
| <b>0</b>  | 8.410000           | 82.50                         | 83.505073                      |
| <b>1</b>  | 5.190000           | 71.18                         | 70.849272                      |
| <b>2</b>  | 5.720000           | 73.25                         | 72.932370                      |
| <b>3</b>  | 8.880000           | 83.64                         | 85.352348                      |
| <b>4</b>  | 5.790000           | 73.64                         | 73.207496                      |
| <b>5</b>  | 8.660000           | 86.99                         | 84.487666                      |
| <b>6</b>  | 7.550000           | 81.18                         | 80.124952                      |
| <b>7</b>  | 7.990000           | 82.75                         | 81.854316                      |
| <b>8</b>  | 7.750000           | 79.50                         | 80.911027                      |
| <b>9</b>  | 8.080000           | 81.70                         | 82.208050                      |
| <b>10</b> | 7.260000           | 79.41                         | 78.985144                      |
| <b>11</b> | 8.750000           | 85.95                         | 84.841400                      |
| <b>12</b> | 6.970000           | 77.19                         | 77.845336                      |
| <b>13</b> | 6.930000           | 78.45                         | 77.688121                      |
| <b>14</b> | 8.340000           | 84.00                         | 83.229947                      |
| <b>15</b> | 8.990000           | 85.46                         | 85.784689                      |
| <b>16</b> | 8.780000           | 84.35                         | 84.959311                      |
| <b>17</b> | 5.640000           | 73.19                         | 72.617940                      |
| <b>18</b> | 7.190000           | 78.21                         | 78.710018                      |
| <b>19</b> | 7.310000           | 77.59                         | 79.181662                      |
| <b>20</b> | 8.610000           | 83.87                         | 84.291147                      |
| <b>21</b> | 8.950000           | 85.15                         | 85.627474                      |
| <b>22</b> | 6.180000           | 72.96                         | 74.740341                      |
| <b>23</b> | 7.860000           | 80.72                         | 81.343368                      |
| <b>24</b> | 5.490000           | 73.61                         | 72.028384                      |
| <b>25</b> | 7.620000           | 79.53                         | 80.400078                      |
| <b>26</b> | 7.260000           | 78.17                         | 78.985144                      |
| <b>27</b> | 8.050000           | 79.63                         | 82.090138                      |
| <b>28</b> | 6.995949           | 76.83                         | 77.947324                      |
| <b>29</b> | 8.090000           | 82.38                         | 82.247353                      |
| <b>30</b> | 6.360000           | 76.04                         | 75.447808                      |
| <b>31</b> | 8.690000           | 85.48                         | 84.605577                      |
| <b>32</b> | 5.390000           | 71.87                         | 71.635346                      |

|    | study_hours | student_marks_original | student_marks_predicted |
|----|-------------|------------------------|-------------------------|
| 33 | 6.370000    | 75.04                  | 75.487112               |
| 34 | 5.050000    | 70.67                  | 70.299020               |
| 35 | 7.260000    | 79.87                  | 78.985144               |
| 36 | 6.330000    | 74.49                  | 75.329897               |
| 37 | 8.670000    | 84.19                  | 84.526970               |
| 38 | 6.010000    | 75.36                  | 74.072178               |
| 39 | 5.340000    | 72.10                  | 71.438828               |

In [ ]:

Fine Tune your Model

In [47]:

lr

Out[47]:

▼ LinearRegression ⓘ ?

- Parameters
- Fitted attributes

◀ ▶

In [48]: lr.score(X\_test, y\_test)

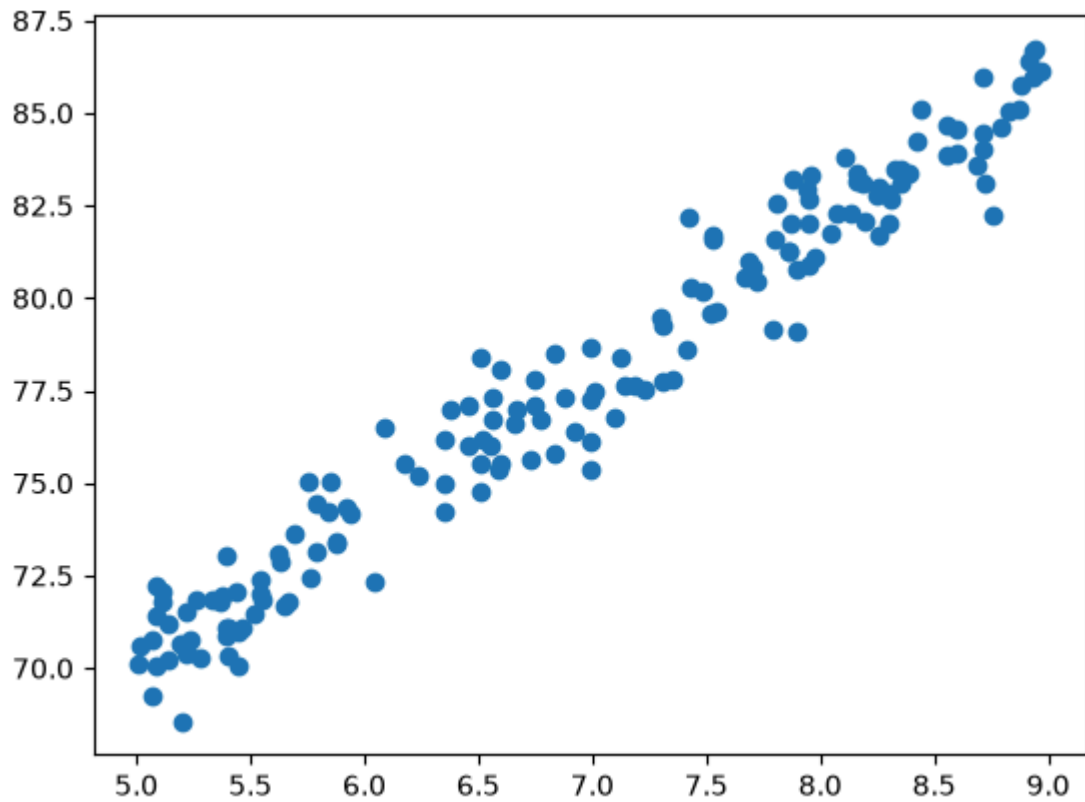
Out[48]: 0.9521841793508594

In [49]: lr.score(X\_test, y\_test)

Out[49]: 0.9521841793508594

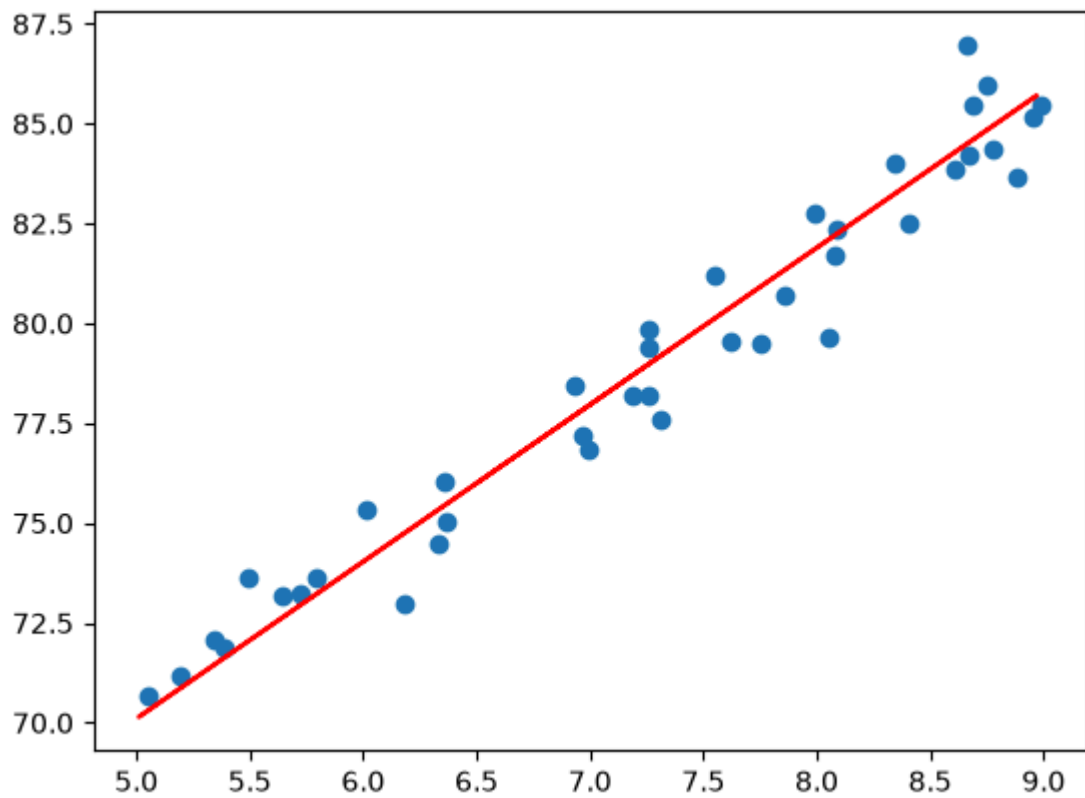
In [50]: plt.scatter(X\_train, y\_train)

Out[50]: <matplotlib.collections.PathCollection at 0x225668a56d0>



```
In [51]: plt.scatter(X_test, y_test)
plt.plot(X_train, lr.predict(X_train), color = "r")
```

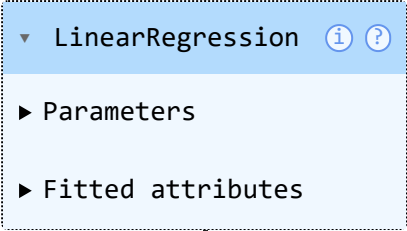
```
Out[51]: [<matplotlib.lines.Line2D at 0x225668f2120>]
```



```
In [ ]:
```

Save ML Model

In [52]: lr

Out[52]:  LinearRegression ⓘ ?

► Parameters

► Fitted attributes

In [53]: `import joblib`  
`joblib.dump(lr, "Desktop.pkl")`

Out[53]: ['Desktop.pkl']

In [54]: `pwd`

Out[54]: 'C:\\\\Users\\Amit Kumar Barnwal'

In [55]: `model = joblib.load("Desktop.pkl")` *#pickle*

In [56]: `pwd`

Out[56]: 'C:\\\\Users\\Amit Kumar Barnwal'

In [ ]: